



---

**UNIVERSITI TEKNOLOGI MARA  
FINAL EXAMINATION**

---

|                    |          |                                        |
|--------------------|----------|----------------------------------------|
| <b>COURSE</b>      | <b>:</b> | <b>DATABASE DESIGN AND DEVELOPMENT</b> |
| <b>COURSE CODE</b> | <b>:</b> | <b>ICT450 / ITS432</b>                 |
| <b>EXAMINATION</b> | <b>:</b> | <b>JULY 2026</b>                       |
| <b>TIME</b>        | <b>:</b> | <b>3 HOURS</b>                         |

**INSTRUCTIONS TO CANDIDATES**

1. This question paper consists of five (5) questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any materials into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
  - i) the Question Paper
  - ii) an Answer Booklet – provided by the Faculty
5. Answer ALL questions in English.

---

**DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO**

---

*This examination paper consists of 5 printed pages.*

---

**QUESTION 1**

- a) Explain the difference between Database and Database Management System (DBMS).

(4 marks)

- b) Aisyah, a new business owner, has launched a startup specializing in "cute and fancy" custom materials, such as waterproof stickers, holographic labels, personalized decorative tapes, and planners. Currently, she keeps all customer, order, and material records in physical folders on a bookshelf, organized by color and alphabetical order. However, she is starting to encounter difficulties as her business expands. Describe **TWO (2)** problems Aisyah faces with her current record-keeping method.

(4 marks)

- c) Briefly explain the **THREE (3)** roles of business rules in the database design process.

(6 marks)

- d) A small car rental company wants to develop a database system to store the company's data. The development team follows the Database Life Cycle (DBLC) to ensure the database is properly developed. Briefly explain **THREE (3)** activities in the first phase of the DBLC that would be involved in developing the database for the company.

(6 marks)

**QUESTION 2**

- a) Object-Oriented Data Model (OODM) is a more recent data model introduced in the database field. Despite its advantages, it is not as widely used as previous data models. Describe **TWO (2)** reasons why OODM is not widely accepted in the database environment.

(4 marks)

- b) A clinic database system contains the following relational schema:

PATIENT (PatientID, Name)  
DOCTOR (DoctorID, DoctorName, DeptID)  
DEPARTMENT (DeptID, DeptName)  
APPOINTMENT (PatientID, DoctorID, DeptID, Date, Time)

The primary key for APPOINTMENT is (PatientID, DoctorID, Date). On several occasions, staff forgot to fill in PatientID in the APPOINTMENT table. In addition, some staff entered DeptID values in the APPOINTMENT table that do not exist in the DEPARTMENT table. Explain all integrity constraints violated and identify the key(s) responsible for enforcing each constraint.

(4 marks)

- c) Differentiate the Top-Down and Bottom-Up design approaches, and explain which approach is better suited for a complex, enterprise-wide database. (6 marks)
- d) Multimedia databases are widely used to preserve cultural heritage. Identify and briefly explain **THREE (3)** examples of multimedia applications used in museums or cultural heritage sites. (6 marks)

### QUESTION 3

Sarah is a hardworking student who keeps track of her continuous assessments using the "Notes" application on her phone. She records the course name, the type of assessment, and the marks she receives. However, as the semester progresses, her notes have become messy and difficult to manage. To stay organized, she decides to design a database called the Student Performance System. Figure 1 shows a sample of her current notes.

Please note that in Sarah's university, each lecturer is assigned to several courses, and each course is assigned to only one lecturer.

| My Performance Record  |           |       |                                  |               |                 |
|----------------------------------------------------------------------------------------------------------|-----------|-------|----------------------------------|---------------|-----------------|
| Assessment Details                                                                                       | Due Date  | Marks | Course Details                   | Lecturer Name | Office Location |
| Q1 - Quiz 1                                                                                              | 11/3/2026 | 35    | CSC402 - Programming 1           | Dr. Hafidzah  | 20, Level 1     |
| Q1 - Quiz 1                                                                                              | 13/3/2026 | 30    | ICT450 - Database Design         | Miss Intan    | 36, Level 2     |
| LQ - Lab Quiz                                                                                            | 20/3/2026 | 51    | ICT450 - Database Design         | Miss Intan    |                 |
| CS - Case Study                                                                                          | 15/4/2026 | 88    | MAT116 - Fundamental Mathematics | Encik Haikal  | 33, Level 2     |
| ASG1 - Assignment 1                                                                                      | 11/3/2026 | 50    | CSC402 - Programming 1           | Dr. Hafidzah  |                 |

**Figure 1:** Sarah's Performance Record

Based on the information given, answer the following questions:

- a) Examine the table shown in Figure 1. Has it been normalized? Identify and explain any problems present in the table. (3 marks)



- b) Identify the primary key(s) and partial dependencies and/or transitive dependencies.  
(6 marks)
- c) Normalize the relational schema in (b) until the third normal form (3NF). Show all your work.  
(11 marks)

#### QUESTION 4

Given the following relational schema for a Fitness Center:

- **TRAINER** (Trainer\_ID, Name, Specialization, Salary)
- **CLASS** (Class\_ID, ClassName, Schedule, Trainer\_ID\*)
- **MEMBER** (Member\_ID, MemberName, Level)
- **ENROLLMENT** (Member\_ID\*, Class\_ID\*, EnrollDate)

Write the SQL statements based on the following requirements:

- a) Create the ENROLLMENT table, including a composite primary key and the necessary foreign keys.  
(4 marks)
- b) Display the names of all trainers who earn more than RM 4,500 and specialize in either 'Yoga' or 'Pilates'. Only display trainers whose names start with the letter 'A'. Sort the results by name in alphabetical order.  
(5 marks)
- c) Display the class names and the total number of members enrolled in each class, but only for classes with more than 15 members who enrolled after 1st January 2026.  
(5 marks)
- d) Using a subquery, display the names of trainers who teach any members currently at the 'Advanced' level. Only include trainers whose salary is greater than RM 4,000.  
(6 marks)

**QUESTION 5**

Every semester, a large number of students apply to stay at the university hostel. Currently, the hostel management filters the applications by manually calculating the total merit points of each student. As the number of applications continues to increase, this process has become inefficient and time-consuming. Therefore, the hostel management has decided to develop a Student Merit Activities Database System to manage student activities and automatically calculate merit points for hostel applications. As a database designer, you are tasked with designing the database system according to the following requirements:

- The system must record student details, including ID, name, phone number, and program code.
- Each student may participate in multiple events, and each event should be participated by at least one student. There is a unique ID for each participation.
- Event details such as ID, name, date, and merit points are recorded.
- An event may require multiple pieces of equipment, and each piece of equipment may be used across multiple events. The system must track the total quantity of equipment used per event.
- Each equipment has its ID, name, and total quantity in store.
- Each event needs to be supervised by one staff member. There are some staff members who do not supervise any events. Finally, each event is led by one student, designated as the Event Leader.

Based on the given information:

- a) Construct a complete Entity Relationship Diagram (ERD) with all of the required tables, attributes, and relationships using the Crow's Foot notation that reflects all of the given business rules.

(16 marks)

- b) Identify **TWO (2)** relevant information that you can get from the ERD you constructed in (a).

(4 marks)

**END OF QUESTION PAPER**